

Lecture 4: Straight Line Complexity of Linear Programs

Algorithms & Geometry of Linear Programming

Daniel Dadush

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Outline

Interior Point Methods, and the Iteration Question

The Central Path

The Max Central Path and Straight Line Complexity

Straight Line Complexity Can Be Exponential

Bounding SLC via Circuits

From simplex to interior point methods

Lectures 2–3: the simplex method — exact, vertex-walking; its practical success explained probabilistically.

Today: **interior point methods** (IPMs) — *unconditional* guarantees.

- *approximate* methods: drive down the optimality gap $c^T x - v^*$
- natural question: *how many iterations to halve the gap?*
- exact optima recovered at the end by rounding: over $\mathbb{Q}$, gap $2^{-\Theta(L)}$ suffices ($L =$ bit size)

Iteration bounds for interior point methods

Standard form, $\mathbf{A} \in \mathbb{R}^{m \times n}$, feasible region of dimension $d = n - m$; one iteration = $O(1)$ linear system solves.

- **[Renegar '88]** log-barrier path following: gap halved every $O(\sqrt{n})$ iterations; exact optimum after $O(\sqrt{n} L)$
- **[Nesterov–Nemirovskii '94]** universal barrier: $O(\sqrt{d})$ per halving
- **[Vavasis–Ye '96]** layered least squares: $O(n^{3.5} \log(\bar{\chi}_{\mathbf{A}} n))$ total, $\bar{\chi}_{\mathbf{A}} \approx \kappa_{\mathbf{A}}$ (Lecture 3) — no dependence on b, c
- **[ABGJ '18]** lower bound: instances where path following takes $2^{\Omega(n)}$ iterations

Which of these bounds reflects reality?

Two mysteries

Mystery 1: practice beats theory

Solvers routinely converge in 20–50 iterations, essentially *regardless of dimension* — far better than $O(\sqrt{n} \log(1/\varepsilon))$.

Mystery 2: theory forbids strong polynomiality [ABGJ '18]

There are instances with $O(n)$ constraints on which every path-following IPM needs $2^{\Omega(n)}$ iterations.

- how can a method with a proven $O(\sqrt{n} L)$ bound need **exponentially many** iterations?
- what distinguishes easy instances from hard ones?

The question, reformulated

Fix only the **paradigm** — *follow the central path* — and ask:

How many iterations does the geometry of the path itself allow?

The answer is a concrete quantity: the **straight line complexity** (SLC) of the central path.

1. **Block 1:** the SLC framework — a lower bound for every path-following method, nearly matched by the method of Lecture 5 [ADLNV '22]
2. **Block 2a:** SLC can be **exponential** [ABGJ '18] — via tropical geometry
3. **Block 2b:** $\text{SLC} \leq \min\{m, n - m\} + 2$ when the circuit imbalance $\kappa_{\mathbf{A}}$ is controlled [DKNOV '24]

Setup: the standard-form pair

$$(P) \quad \min c^\top x \text{ s.t. } \mathbf{A}x = b, x \geq \mathbf{0} \qquad (D) \quad \max b^\top y \text{ s.t. } \mathbf{A}^\top y + s = c, s \geq \mathbf{0}$$

- $\mathbf{A} \in \mathbb{R}^{m \times n}$ full row rank; both sides **strictly** feasible; $v^* :=$ common optimal value; (x^*, s^*, y^*) optimal
- **gap formula** (Lecture 1): for feasible $x, (s, y)$: $c^\top x - b^\top y = s^\top x \geq 0$
- in particular $s^{*\top} x = c^\top x - v^*$: a **nonnegative** functional vanishing exactly on the optimal face
- **gap-truncated regions** (bounded, by dual strict feasibility):

$$\mathcal{P}_g := \{x : \mathbf{A}x = b, x \geq \mathbf{0}, c^\top x \leq v^* + g\}, \quad g \geq 0$$

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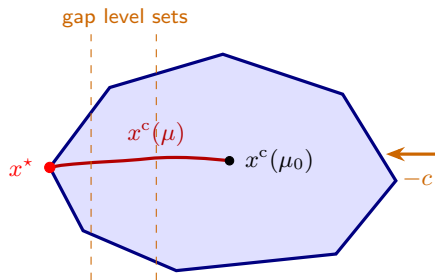
The central path

Definition (Central path)

For $\mu > 0$:

$$x^c(\mu) := \operatorname{argmin}_{\mathbf{A}x=b, x>0} c^T x - \mu \sum_{i=1}^n \ln x_i$$

- the **log-barrier** blows up at the boundary: interior minimizer, unique
- μ large: near the “center”; $\mu \rightarrow 0$: $x^c(\mu) \rightarrow$ optimal x^*
- a **path-following IPM** tracks the path as $\mu \rightarrow 0$ (Lecture 5)



The path nearly maximizes every coordinate

Lemma

Let $\mu > 0$ and let $x' \in \mathcal{P}$ satisfy $c^\top x' \leq c^\top x^c(\mu)$. Then $x'_i \leq n x_i^c(\mu)$ for every $i \in [n]$.

Proof. Write $x := x^c(\mu)$: an interior minimizer of $f(w) := c^\top w - \mu \sum_i \ln w_i$ over $\{w : \mathbf{A}w = b\}$, so $\nabla f(x) = c - \mu x^{-1} \perp \ker(\mathbf{A}) \ni x' - x$. Hence

$$0 = (c - \mu x^{-1})^\top (x' - x) = \underbrace{c^\top x' - c^\top x}_{\leq 0} - \mu \sum_{i=1}^n \frac{x'_i}{x_i} + \mu n \implies \sum_{i=1}^n \frac{x'_i}{x_i} \leq n.$$

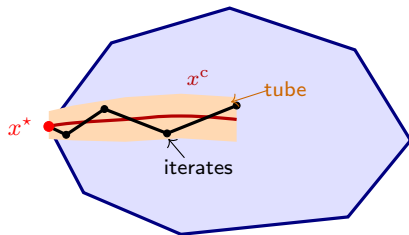
Every term is nonnegative, so each single term is at most n . □

Path-following methods, abstractly

Whatever their internal details, path-following IPMs share one behavior:

- iterates stay **multiplicatively close** to the path: each coordinate within a factor γ of the path point with the same gap
- consecutive iterates joined by segments inside this multiplicative **tube**
- $\Rightarrow$ a T -iteration run traces a **piecewise linear curve with T pieces** inside the tube

Lower bounding T is now geometric: *how few pieces can such a curve have?*



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The max central path

Definition (Max central path)

For $i \in [n]$ and $g \geq 0$: $x_i^{\text{m}}(g) := \max\{x_i : x \in \mathcal{P}_g\}$, the largest i -th coordinate at gap g .

- each x_i^{m} is **nondecreasing, concave, piecewise linear**
- the vector $x^{\text{m}}(g)$ is generally *not* feasible — one concave curve per coordinate

Corollary (Sandwich) — write $g_\mu := c^\top x^{\text{c}}(\mu) - v^*$

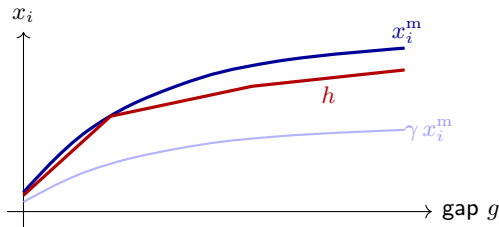
$$\frac{1}{n} x_i^{\text{m}}(g_\mu) \leq x_i^{\text{c}}(\mu) \leq x_i^{\text{m}}(g_\mu) \quad \text{for all } \mu > 0, i \in [n]$$

Upper bound: $x^{\text{c}}(\mu) \in \mathcal{P}_{g_\mu}$. Lower bound: the Lemma, applied to the maximizer attaining $x_i^{\text{m}}(g_\mu)$.

Straight line complexity

Definition (Straight line complexity)

For $f : [0, g_0] \rightarrow \mathbb{R}_+$ and $\gamma \in (0, 1]$, $\text{SLC}_\gamma(f, g_0) :=$ minimum number of pieces of a continuous piecewise linear $h : [0, g_0] \rightarrow \mathbb{R}_+$ with $\gamma f \leq h \leq f$ on $[0, g_0]$.



Here three pieces suffice: $\text{SLC}_\gamma(x_i^m, g_0) \leq 3$.

SLC lower-bounds every path-following method

Proposition

Let $x(t)$ be a piecewise linear curve in $\mathcal{P}$ with T pieces, its gap strictly decreasing from g_0 to 0, satisfying the tube condition $\gamma x_i^c \leq x_i(t) \leq \gamma^{-1} x_i^c$ (at matching gaps, for all i). Then

$$T \geq \max_{i \in [n]} \text{SLC}_{\gamma^2/n}(x_i^m, g_0).$$

Proof. Fix i . On each segment the gap is affine in t and strictly decreasing, so coordinate i as a function of the gap is piecewise linear with at most T pieces: call it $\tilde{h}(g)$. Tube condition + sandwich:

$$\frac{\gamma}{n} x_i^m \leq \gamma x_i^c \leq \tilde{h} \leq \gamma^{-1} x_i^c \leq \gamma^{-1} x_i^m,$$

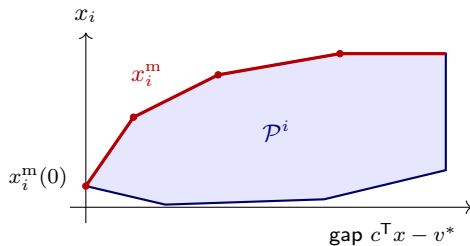
so $h := \gamma \tilde{h}$ satisfies $(\gamma^2/n) x_i^m \leq h \leq x_i^m$. □

The max central path is a shadow

Project $\mathcal{P}$ onto the plane of the gap and coordinate i :

$$\mathcal{P}^i := \{(c^\top x - v^*, x_i) : x \in \mathcal{P}\}$$

- graph of x_i^m = increasing part of the **upper boundary** of $\mathcal{P}^i$
- breakpoints = projected **vertices**,
pieces = projected **edges**: a **shadow vertex simplex path** (Lectures 2–3)
- unconditional — if enormous —
bound: $\text{SLC}_1(x_i^m, g_0) \leq \binom{n}{m} + 1$



Interior point methods match the geometry of the path

Theorem (ADLNV '22)

There is a path-following IPM (the TRUST-REGION IPM of Lecture 5) which, started near the central path at gap g_0 , computes an exact optimal primal-dual pair within

$$O\left(\min_{\gamma \in (0,1]} n^{1.5} \log\left(\frac{n}{\gamma}\right) \sum_{i=1}^n \text{SLC}_{\gamma}(x_i^{\text{m}}, g_0)\right) \text{ iterations.}$$

- with the Proposition: SLC is the **floor** for every path-following method and, up to $\tilde{O}(n^{2.5})$, the **ceiling** for the best one
- iteration complexity of path following = geometry of the max central path

The rest of the lecture computes this quantity.

A benign example: n long straight stretches

For $0 < \varepsilon \ll 1$:

$$\min \sum_{i=1}^n \varepsilon^{i-1} x_i \text{ s.t. } \sum_{i=1}^n x_i = 1, \quad x \geq \mathbf{0}.$$

- the central path descends through n long, nearly straight segments, visiting the neighborhoods of the vertices $e_n, e_{n-1}, \dots, e_1$ in order
- yet each x_i^m has $O(1)$ pieces, so $\sum_i \text{SLC}_1(x_i^m) = O(n)$ (problem session)
- by [ADLNV '22]: solved in $\tilde{O}(n^{2.5})$ iterations — visiting many vertices is harmless as long as the path is **straight**

Where we are

Block 1 — the SLC framework:

- sandwich: $\frac{1}{n}x_i^m \leq x_i^c \leq x_i^m$ coordinatewise
- $\max_i \text{SLC}_{\Omega(1/n)}(x_i^m, g_0)$ **lower-bounds** every path-following IPM
- some path-following IPM **matches** $\sum_i \text{SLC}$ up to $\tilde{O}(n^{1.5})$ factors [ADLNV '22]

Block 2 — computing the SLC:

1. can it be large? **Yes**: $2^n - 1$ on explicit instances, via *tropical geometry* [ABGJ '18]
2. what keeps it small? **Circuits**: $\text{SLC} \leq \min\{m, n - m\} + 2$ when κ_A is bounded [DKNOV '24]

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An explicit hard instance

For a parameter $t > 1$, define $\text{LP}(t)$ in variables $x_1, \dots, x_{2n+1}$:

$$\begin{array}{ll}\min & x_1 \\ \text{s.t.} & x_1 \leq t^2, \quad x_2 \leq t \\ & x_{2j+1} \leq t x_{2j-1}, \quad x_{2j+1} \leq t x_{2j}, \quad 1 \leq j \leq n \\ & x_{2j+2} \leq t^{1-1/2^j} (x_{2j-1} + x_{2j}), \quad 1 \leq j < n \\ & x_{2n-1} \geq 0, \quad x_{2n} \geq 0\end{array}$$

a chain: each pair bounded by the previous pair; optimal value 0, so the gap of a feasible x is x_1

Theorem (ABGJ '18)

For every $\gamma \in (0, 1)$ and $t \geq 2^{\Omega(2^n n \log(n/\gamma))}$: $\text{SLC}_\gamma(x_{2n}^{\text{m}}) \geq 2^n - 1$.

Reading the statement, and the plan

- $LP(t)$ has $O(n)$ constraints — but t is **doubly exponential**, so the bit size $L \approx \log t$ is exponential
- consistent with Renegar's $O(\sqrt{n} L)$: the theorem separates **strongly** from **weakly** polynomial exactly

Proof strategy: pass to the tropical limit

Substitute $x_j = t^{y_j}$ and work on the $\log_t$ scale: for enormous t , the LP degenerates to a piecewise linear — **tropical** — problem.

1. **the tropical shadow**: the tropical max central path is an explicit **staircase** with 2^n steps
2. **the collapse**: the LP max central path is glued to the staircase
3. **tracking argument**: a piecewise linear curve needs one piece per staircase step

Step 1: the tropical shadow

On the $\log_t$ scale, multiplication becomes addition, and addition becomes maximization up to the **softmax inequalities**

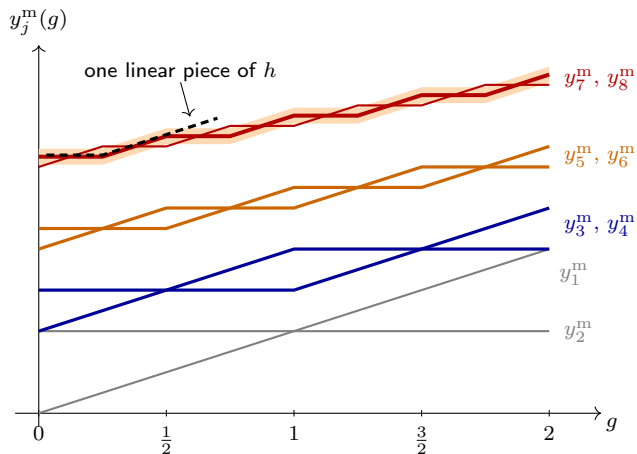
$$\max\{a, b\} \leq \log_t (t^a + t^b) \leq \max\{a, b\} + \log_t 2.$$

Taking $\log_t$ of the constraints gives, up to error $\log_t 2$, the **tropical program** ($y_1 = \text{gap}$ on the $\log_t$ scale):

$$y_1 \leq 2, \quad y_2 \leq 1, \quad y_{2j+1} \leq 1 + \min\{y_{2j-1}, y_{2j}\}, \quad y_{2j+2} \leq 1 - \frac{1}{2^j} + \max\{y_{2j-1}, y_{2j}\}$$

- its max central path $y_j^m(g)$ ($= \max y_j$ s.t. $y_1 \leq g$) unrolls by an explicit recursion
- the max-constraint lets y_{2j+2} inherit the *better* predecessor; the shrinking offsets $1/2^j$ alternate which one is better as g varies: breakpoints **double at every level**
 - y_3^m, y_4^m have 1; y_5^m, y_6^m have 3; y_7^m, y_8^m have 7; ...

The staircase cascade ($n = 4$)



Every level interleaves and doubles the breakpoints of the level below; the top curve $y_{2^n}^m$ is a staircase with 2^n geometrically shrinking steps.

Step 2: the collapse

The softmax errors accumulate along the chain — but only **linearly**:

Collapse estimate (induction through the j layers of constraints)

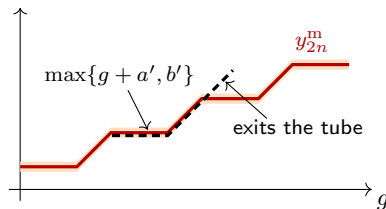
$$\left| \log_t x_j^{\text{m}}(t^g) - y_j^{\text{m}}(g) \right| \leq j \log_t 2, \quad g \in [0, 1]$$

- for t doubly exponential in n : total error $2n \log_t 2 \ll 2^{-n}$ — a small fraction of a *single* staircase step
- so on the $\log_t$ scale, the LP max central path is **glued** to the tropical staircase: it lives in the narrow shaded band of the figure

Step 3: tracking the staircase

Let h be piecewise linear with $\gamma x_{2n}^m \leq h \leq x_{2n}^m$;
set $H(g) := \log_t h(t^g)$.

- by Step 2, H lies in an **additive** tube of half-width $\delta \ll 2^{-n}$ around the staircase
- a linear piece $h(\xi) = a\xi + b$ becomes $H(g) \in \max\{g + \log_t a, \log_t b\} + [0, \log_t 2]$: constant, then slope one, with **one** transition
- so each piece tracks the alternating flat runs and rises for $\leq \approx 2$ steps



Tracking all 2^n steps requires exponentially many pieces; a careful count gives
 $\text{SLC}_\gamma(x_{2n}^m) \geq 2^n - 1$. [ABGJ '18]

Consequences of the lower bound

Combining with the SLC framework of Block 1:

- **no** path-following IPM terminates in fewer than $2^n - 1$ iterations on $\text{LP}(t)$
- in particular, **no path-following IPM is strongly polynomial**
- by [ADLNV '22], the trust-region IPM needs at most $\tilde{O}(n^{2.5} 2^n)$ iterations here: floor and ceiling are *both* exponential — the SLC characterization is tight

The circuit imbalance κ of $\text{LP}(t)$ grows with t (problem session) — next: bounded κ forces small SLC.

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Circuits and circuit imbalance (Lecture 3 recap)

- **circuits** of $\mathbf{A}$: support-minimal nonzero vectors of $\ker(\mathbf{A})$, unique up to scaling
- **conformal circuit decomposition**: every $w \in \ker(\mathbf{A})$ is a sum $w = \sum_{j=1}^{\ell} h^{(j)}$ of $\ell \leq n$ circuits with the signs of w — partial sums stay in the box $[\mathbf{0}, w]$
- **circuit imbalance**: $\kappa_{\mathbf{A}} := \max \left\{ |h_j/h_k| : h \text{ circuit}, j, k \in \text{supp}(h) \right\}$
- $\kappa_{\mathbf{A}} = 1$ for network (and all totally unimodular) matrices; Δ -modular integer matrices: $\kappa_{\mathbf{A}} \leq \Delta$

The upper bound theorem, and preprocessing

Theorem (DKNOV '24)

With $\kappa := \kappa_{\mathbf{A}}$: $\text{SLC}_{1/(n^2\kappa^2)}(x_i^{\mathbf{m}}) \leq \min\{m, n - m\} + 2$ for every $i \in [n]$.

- uniform in g_0 , independent of b and c : **only the constraint matrix enters**
- **preprocessing**: fix a *basic* optimal x^* attaining $x_i^* = x_i^{\mathbf{m}}(0)$ and a basic optimal dual (s^*, y^*) ; complementary slackness:

$$|\text{supp}(x^*)| \leq m, \quad |\text{supp}(s^*)| \leq n - m, \quad \text{supp}(x^*) \cap \text{supp}(s^*) = \emptyset$$

- gap formula: membership in $\mathcal{P}_g$ reads $s^{*\top}x \leq g$

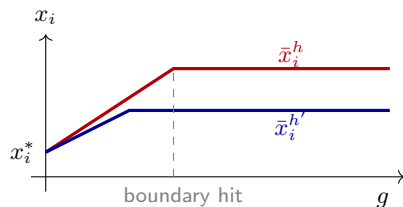
Plan: approximate $x_i^{\mathbf{m}}$ by a few elementary curves — one per circuit in a small *cover*.

Circuit curves

For $h \in \ker(\mathbf{A})$, the **h -curve**
 $\bar{x}^h(g) := x^* + \alpha^h(g) h$ is the farthest point of
 $\mathcal{P}_g$ reachable from x^* along h :

$$\alpha^h(g) = \min \left(\underbrace{\frac{g}{s^{*\top} h}}_{\text{gap budget}}, \underbrace{\min_{j: h_j < 0} \frac{x_j^*}{|h_j|}}_{\text{first blocking coord.}} \right)$$

$\bar{x}_i^h(g)$ is concave with **two pieces**: rise from x_i^*
at slope $h_i/(s^{*\top} h)$, then constant once the
moving point hits the boundary



Circuit curves capture the max central path

Lemma

For every $g \geq 0$ with $x_i^{\text{m}}(g) > x_i^$ there is a scaled circuit h with $h_i > 0$ and $\bar{x}_i^h(g) \geq x_i^{\text{m}}(g)/n$.*

Proof. Let $\hat{x} \in \mathcal{P}_g$ attain $x_i^{\text{m}}(g)$; conformally decompose $\hat{x} - x^* = \sum_{j=1}^{\ell} h^{(j)}$, $\ell \leq n$.

- conformality: each $x^* + h^{(j)}$ stays in the box between x^* and $\hat{x} \Rightarrow$ feasible; optimality of x^* gives $s^{*\text{T}}h^{(j)} = c^{\text{T}}h^{(j)} \geq 0$, and $s^{*\text{T}}h^{(j)} \leq s^{*\text{T}}(\hat{x} - x^*) \leq g$
- hence $x^* + h^{(j)} \in \mathcal{P}_g$, i.e., $\alpha^{h^{(j)}}(g) \geq 1$; choose k maximizing $h_i^{(k)}$:

$$\bar{x}_i^{h^{(k)}}(g) \geq x_i^* + h_i^{(k)} \geq x_i^* + \frac{1}{\ell} \sum_j h_i^{(j)} \geq \hat{x}_i / \ell \geq x_i^{\text{m}}(g) / n \quad \square$$

Circuit covers

Definition (γ -circuit cover)

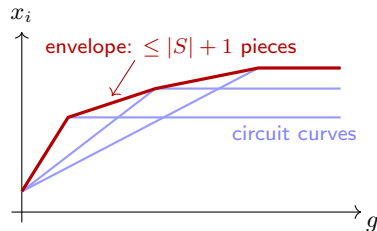
$S \subseteq \ker(\mathbf{A})$ (finite, $c^\top h \geq 0$ on S) is a γ -circuit cover of i if every scaled circuit h' with $c^\top h' \geq 0$ is dominated: some $h \in S$ has $\bar{x}_i^h \geq \gamma \bar{x}_i^{h'}$ pointwise.

Lemma (covers bound SLC)

S a γ -circuit cover of $i \Rightarrow$

$$\text{SLC}_{\gamma/n}(x_i^m) \leq |S| + 1.$$

- the **upper concave envelope** of $|S|$ two-piece curves has $\leq |S| + 1$ pieces
- capture lemma + cover property: the envelope fits in the (γ/n) -tube of x_i^m



It remains to construct a small cover — this is where κ enters.

A small cover from the circuit imbalance

Theorem (DKNOV '24)

For every i there is a $(1/(n\kappa^2))$ -circuit cover of i of size at most $\min\{m, n - m\} + 1$.

Construction. Order coordinates so that $x_1^* \geq \dots \geq x_n^*$ and $s_1^* \leq \dots \leq s_n^*$ (disjoint supports); $p := |\text{supp}(x^*)| \leq m$, $\text{supp}(s^*) \subseteq \{d, \dots, n\}$, $n - d + 1 \leq n - m$.

- **relevant circuits:** normalize $h_i = 1$; discard h with $h_j < 0$ for some $j \notin \text{supp}(x^*)$ (their curves never move); note $1/\kappa \leq |h_j| \leq \kappa$ on $\text{supp}(h)$
- **signature:** $I^p(h) := \max\{j \leq p : h_j < 0\}$ (deepest **blocking** coordinate),
 $I^d(h) := \max\{j \geq d : h_j > 0\}$ (most expensive **paying** coordinate); $\max \emptyset := 0$
- $S :=$ one representative per **Pareto-minimal** signature $\Rightarrow$
 $|S| \leq \min\{p, n - d + 1\} + 1 \leq \min\{m, n - m\} + 1$

Why the cover works: slopes and caps

A two-piece curve is determined by its **slope** $1/(s^{*\top}h)$ and its **cap** $\min_{j:h_j<0} x_j^*/|h_j|$ (recall $h_i = 1$). Given a relevant circuit h , take $h' \in S$ with $I^p(h') \leq I^p(h)$, $I^d(h') \leq I^d(h)$.

- **slopes** ($j' := I^d(h') \leq j'' := I^d(h)$; s^* increasing; entries in $[1/\kappa, \kappa]$):

$$s^{*\top}h' \leq n\kappa s_{j'}^* \leq n\kappa s_{j''}^* \leq n\kappa^2 s_{j''}^* h_{j''} \leq n\kappa^2 s^{*\top}h$$

$\Rightarrow$ slope of the h' -curve $\geq \frac{1}{n\kappa^2} \times$ slope of the h -curve

- **caps** (x^* decreasing; the blocking coordinate of h' is *shallower*, hence larger):

$$\text{cap}(h') \geq \frac{x_{I^p(h)}^*}{\kappa} \geq \frac{1}{\kappa^2} \text{cap}(h)$$

- slope and cap within these factors $\Rightarrow$ pointwise domination up to $1/(n\kappa^2)$ (details: notes)

Consequences: strongly polynomial territory for IPMs

Plugging into [ADLNV '22]: the trust-region IPM solves *any* LP within

$$O\left(n^{1.5} \min\{m, n - m\} \log(n \kappa_{\mathbf{A}})\right) \text{ iterations}$$

— a bound depending on the **constraint matrix alone**.

- bounded $\kappa_{\mathbf{A}}$ (network, Δ -modular): **strongly polynomial** iteration bounds via interior point methods
- caveat: a well-centered start is required, and standard initialization does not preserve κ — a κ -safe initialization needs further ideas [DKNOV '24]
- refinement: ≤ 2 nonzeros per column $\Rightarrow \text{SLC}_{1/(mn)}(x_i^m) = O(mn)$ — the strongly polynomial algorithm from the Lecture 2 landscape

Summary and what comes next

Today: straight line complexity is both floor and ceiling for path-following interior point methods:

- no path-following method beats $\max_i \text{SLC}(x_i^m)$ iterations
- explicit instances where this is $2^n - 1$ [ABGJ '18]: no path-following IPM is strongly polynomial
- $\text{SLC}(x_i^m) \leq \min\{m, n - m\} + 2$ when κ_A is bounded [DKNOV '24]

Tomorrow (Lecture 5): the missing piece — the algorithm achieving the matching upper bound: predictor-corrector IPMs, the classical $O(\sqrt{n} \log(1/\varepsilon))$ analysis, and the **trust-region step** [ADLNV '22].

Problem session: the tropical staircase for $n = 2$; max central path duality; the ε -simplex path; the circuit imbalance of $\text{LP}(t)$.